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Brain Tumor Segmentation Using U-Net

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Introduction

- **Medical Challenge:**
 - Brain tumors are dangerous neurological disorders that require accurate and early diagnosis for effective treatment.
- **Traditional Limitation:**
 - Manual analysis and segmentation of MRI scans is time consuming and may lead to human error.
- **Deep Learning Solution:**
 - Artificial Intelligence and Deep Learning techniques help automate tumor detection and segmentation from MRI images.
- **Why U-Net?**
 - U-Net is a Convolutional Neural Network (CNN) specially designed for biomedical image segmentation and provides high accuracy in detecting tumor regions.

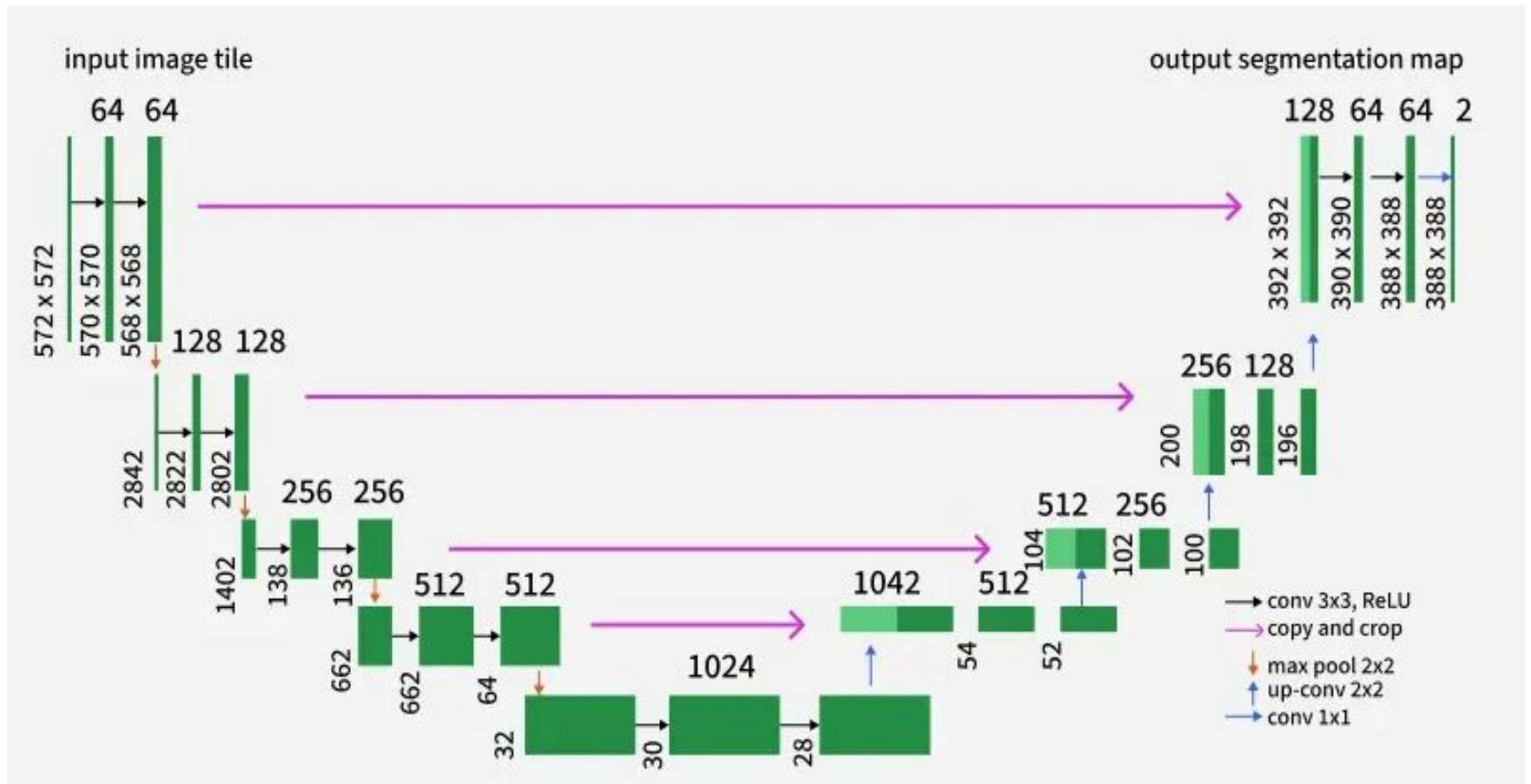
Objectives

- To develop a deep learning based system for automatic brain tumor segmentation.
- To preprocess MRI brain scan images for efficient model training.
- To implement U-Net architecture for medical image segmentation.
- To evaluate model performance using segmentation metrics.
- To reduce manual effort involved in medical image analysis.

Literature Survey

Article	Approach/Method	Advantages	Limitations	Remarks
"U-Net: Convolutional Networks for Biomedical Image Segmentation" by Olaf Ronneberger (2015) [1]	Encoder–decoder CNN architecture with skip connections for medical image segmentation	High segmentation accuracy, works well on small datasets	May lose fine details in complex tumors	Foundation model for many medical segmentation networks
"Brain Tumor Segmentation Using U-Net Based Deep Learning Architecture" by Mehmet Seker (2020) [2]	2D U-Net applied slice-by-slice on MRI volumes for glioma segmentation	Simple implementation, effective on BraTS dataset	Does not exploit 3D spatial context between slices	Demonstrates U-Net's strong baseline performance on brain MRI
"Multi-Scale U-Net for Brain Tumor Segmentation on MRI Images" by Deng et al. (2022) [3]	U-Net with multi-scale feature extraction using dilated convolutions	Captures both local and global tumor features	More hyperparameters to tune	Improves segmentation of heterogeneous tumor regions

Architecture



Architecture

Input

- MRI image of size **572×572**
- Image is fed into the U-Net model for tumor segmentation.

1. Contracting Path (Encoder)

- Consists of repeated:
 - **3×3 Convolution + ReLU**
 - **2×2 Max Pooling**
- Extracts important features such as edges, textures, and tumor patterns.
- Spatial size decreases while feature channels increase.

Feature Channels

$64 \rightarrow 128 \rightarrow 256 \rightarrow 512 \rightarrow 1024$

2. Bottleneck Layer

- Deepest layer of the network.
- Contains **1024 feature maps**.
- Learns high-level tumor representations.

3. Expanding Path (Decoder)

- Uses **Up-Convolution (Transpose Convolution)** for upsampling.
- Restores original image resolution.
- Combines encoder and decoder features using **Skip Connections**.

Feature Channels

$1024 \rightarrow 512 \rightarrow 256 \rightarrow 128 \rightarrow 64$

4. Skip Connections

- Transfer feature maps from encoder to decoder.
- Preserve fine spatial details lost during pooling.
- Improve localization and segmentation accuracy.

5. Output Layer

- Final **1×1 Convolution** generates segmentation map.
- Output size: **$388 \times 388 \times 2$**

Dataset Used

Dataset (BraTS2020)

- **369 patients** with brain tumor MRI scans
- Each patient has **4 MRI modalities**:

Modality	What it Shows
T1	Basic brain structure
T1CE	Tumor enhanced with contrast dye
T2	Fluid content in brain
FLAIR	Lesions, suppresses fluid signals

- **Tumor Labels:**
 - Label 0 – Background (Not Tumor)
 - Label 1 – Necrotic/Non-enhancing Core
 - Label 2 – Peritumoral Edema
 - Label 4 – GD-Enhancing Tumor (originally label 4)

Tools & Technologies

- **Programming Language:** Python
- **Libraries:**
 - **TensorFlow / Keras:** For U-Net model implementation and training.
 - **NumPy:** For numerical computations and array processing.
 - **OpenCV:** For image preprocessing and segmentation operations.
 - **Matplotlib:** For visualization of MRI images and training results.
 - **Scikit-learn:** For preprocessing and data handling.
 - **Nibabel:** For loading and processing MRI **.nii** files.
- **Model Architecture:** U-Net Convolutional Neural Network (CNN)
- **Platform:** Google Colab with GPU acceleration.

Implementation Steps

1. **Dataset Collection**

- Collect MRI brain images and tumor masks.

2. **Preprocessing**

- Resize and normalize images.
- Split into training and testing datasets.

3. **Data Augmentation**

- Apply rotation, flipping, and zooming.

4. **Build U-Net Model**

- Encoder → Feature Extraction
- Bottleneck → Deep Features
- Decoder → Segmentation Output

5. **Model Compilation**

- Optimizer: Adam
- Loss: Dice/Binary Cross Entropy

Implementation Steps

6. **Model Training**
 - Train U-Net using MRI images and masks.
7. **Tumor Segmentation**
 - Predict tumor region pixel-by-pixel.
8. **Post-Processing**
 - Remove noise and refine boundaries.
9. **Performance Evaluation**
 - Metrics: Accuracy, Dice Score, IoU.
10. **Final Output**
 - Segmented MRI image highlighting tumor area.

1. Multi-Modal Brain MRI Images

The figure shows different MRI modalities used for brain tumor analysis:

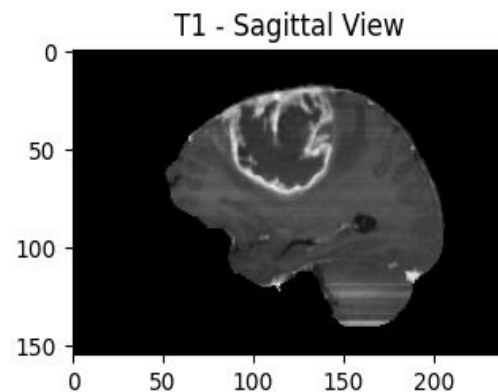
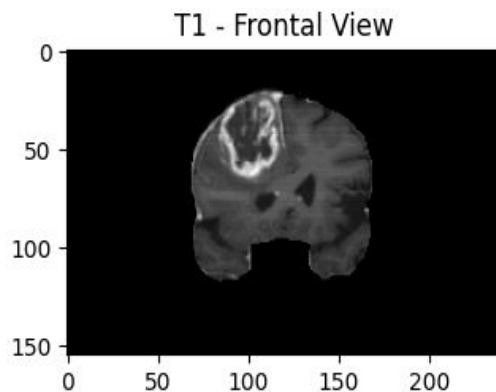
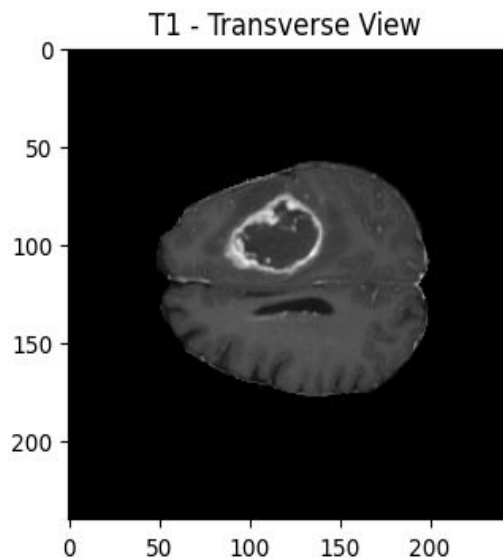
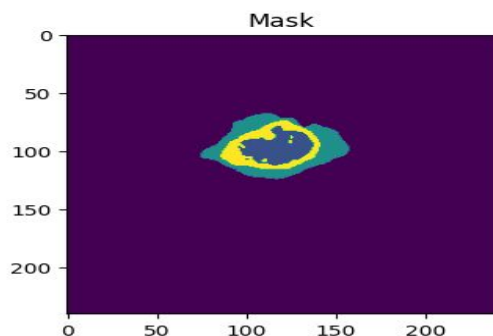
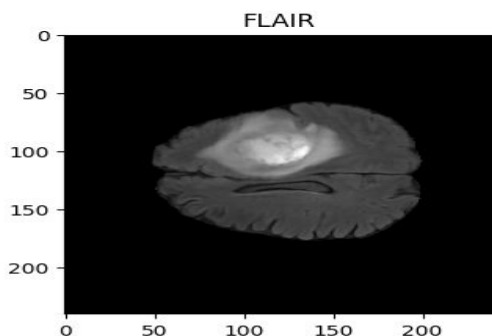
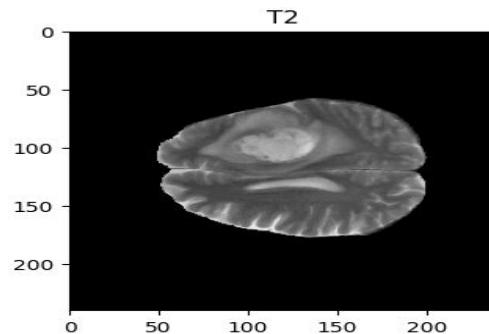
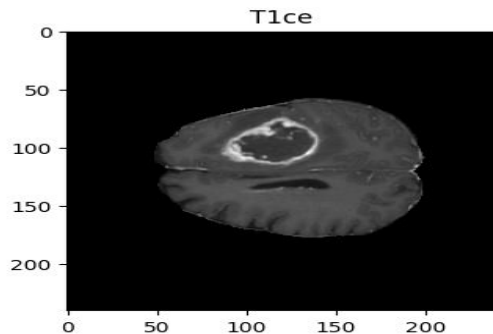
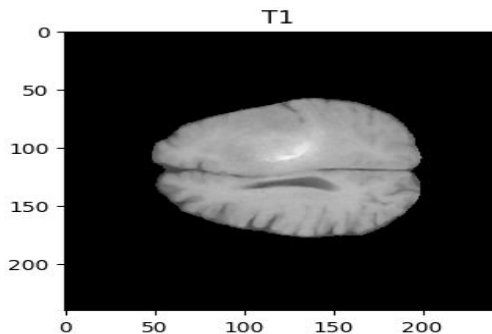
- **T1 Image** – Provides anatomical structure of the brain.
- **T1ce Image** – Contrast-enhanced MRI highlighting tumor boundaries clearly.
- **T2 Image** – Displays fluid and edema regions around the tumor.
- **FLAIR Image** – Suppresses fluid signals and enhances abnormal tumor regions.
- **Mask Image** – Ground truth segmentation mask indicating tumor regions used for supervised learning.

2. Different Views of Brain MRI

The lower figure shows the tumor from different orientations:

- **Transverse View** – Horizontal cross-sectional view of the brain.
- **Frontal (Coronal) View** – Front view showing left and right hemispheres.
- **Sagittal View** – Side view representing depth and tumor position.

Results



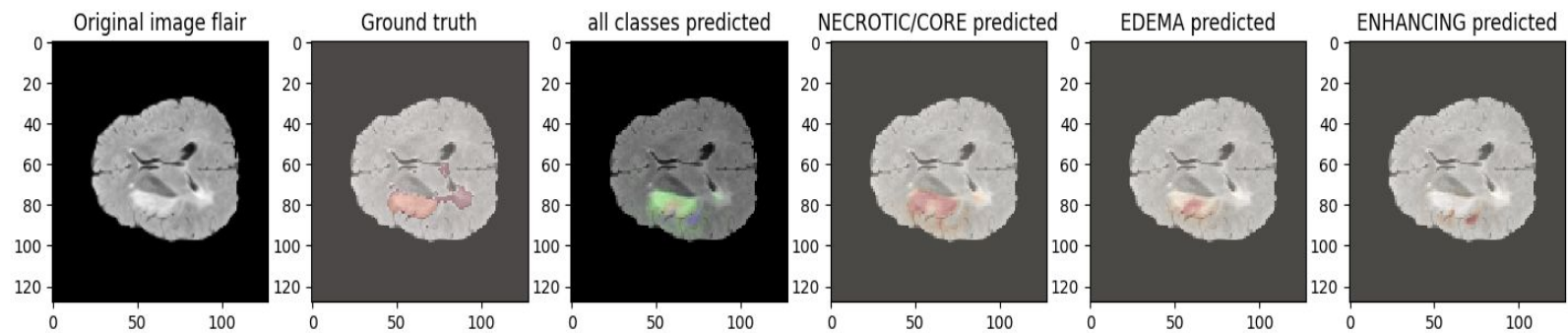


fig : Brain Tumor Segmentation Output Using U-Net

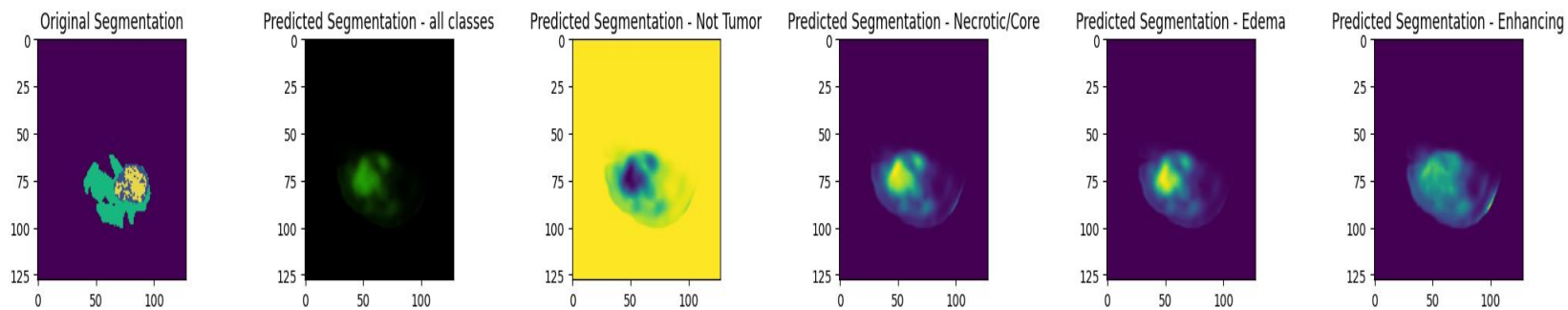
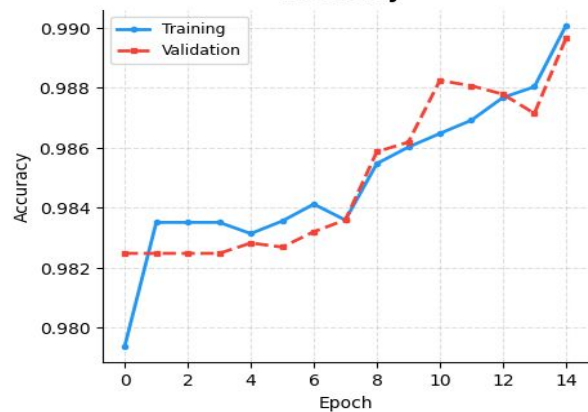


fig : Predicted Multi-Class Brain Tumor Segmentation Results Using U-Net

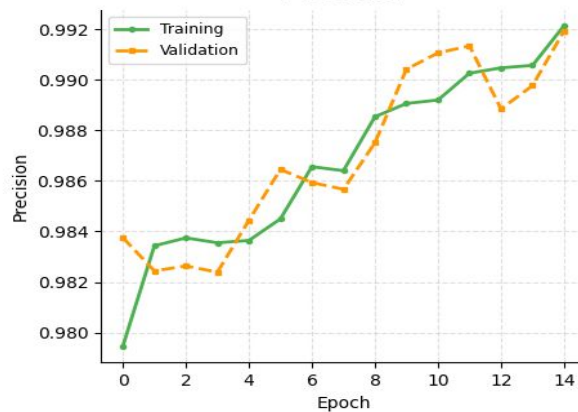
Accuracy Graph

Brain Tumor U-Net - Complete Training Metrics Dashboard

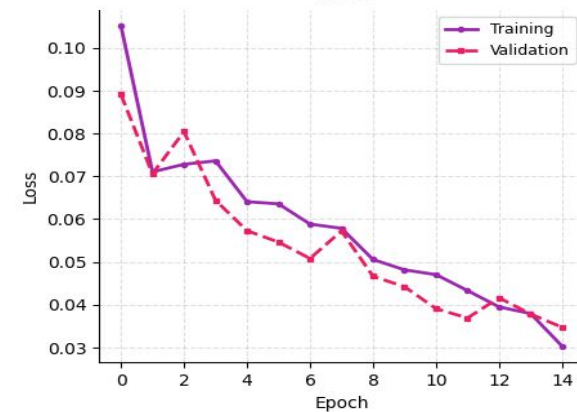
Accuracy



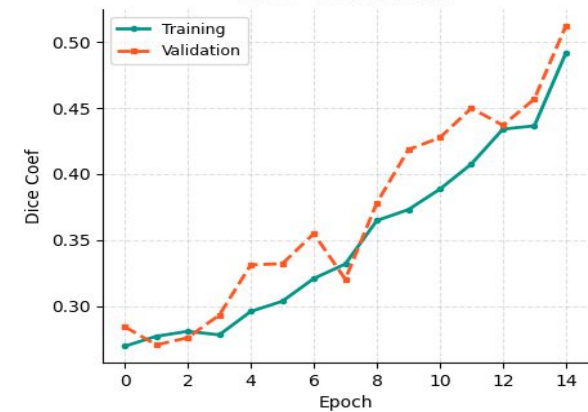
Precision



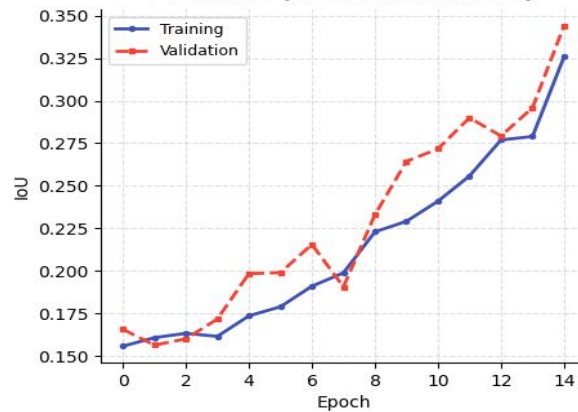
Loss



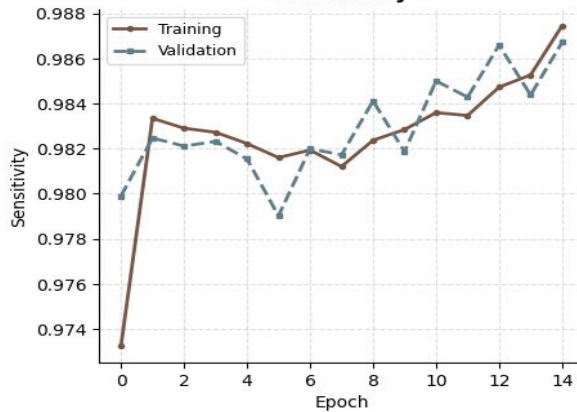
Dice Coefficient



Mean IoU (derived from Dice)



Sensitivity



Advantages

- **High Segmentation Accuracy**
 - Provides precise pixel-level tumor detection.
- **Better Tumor Localization**
 - Skip connections preserve important spatial details.
- **Works Well on Medical Images**
 - Designed specifically for biomedical image segmentation.
- **Efficient with Limited Data**
 - Achieves good performance even with smaller datasets.
- **Reduces Manual Effort**
 - Helps doctors in faster and more accurate diagnosis.

Future Scope

1. **Improve Segmentation Accuracy**

- Use advanced architectures like Attention U-Net or U-Net++.

2. **Real-Time Tumor Detection**

- Develop faster models for real-time medical diagnosis.

3. **3D Brain Tumor Segmentation**

- Extend the model to process 3D MRI volumes for better analysis.

4. **Multi-Class Tumor Classification**

- Segment different tumor regions such as edema, necrosis, and enhancing tumor separately.

5. **Clinical Integration**

- Integrate the system into hospital diagnostic tools to assist radiologists and doctors.

Conclusion / Q&A

- The proposed U-Net model effectively performs brain tumor segmentation from MRI images.
- The encoder-decoder architecture with skip connections helps achieve accurate tumor localization and boundary detection.
- Multi-modal MRI images improve feature extraction and segmentation performance.
- The model reduces manual effort and assists doctors in faster diagnosis and treatment planning.
- Overall, U-Net provides an efficient and reliable solution for medical image segmentation applications.

References

- [1] O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional Networks for Biomedical Image Segmentation,” Proc. MICCAI, pp. 234–241, 2015.
- [2] Z. Zhou, M. M. R. Siddiquee, N. Tajbakhsh, and J. Liang, “UNet++: A Nested U-Net Architecture for Medical Image Segmentation,” Deep Learning in Medical Image Analysis, 2018.
- [3] V. Badrinarayanan, A. Kendall, and R. Cipolla, “SegNet: A Deep Convolutional Encoder-Decoder Architecture for Image Segmentation,” IEEE Trans. Pattern Anal. Mach. Intell., vol. 39, no. 12, pp. 2481–2495, 2017.
- [4] B. H. Menze et al., “The Multimodal Brain Tumor Image Segmentation Benchmark (BraTS),” IEEE Trans. Med. Imaging, vol. 34, no. 10, pp. 1993–2024, 2015.
- [5] S. Bakas et al., “Advancing The Cancer Genome Atlas Glioma MRI Collections with Expert Segmentation Labels,” Sci. Data, vol. 4, 2017.

Thank You